Roll No.

Amrita Vishwa Vidyapeetham

Mid Term Examinations October 2025

Third Semester

Int. MSc Physics

24PHY205 Intermediate Mechanics 1

Duration 2 hours | Max : 50 marks

Answer all questions

Section A : 5 marks each

1. A barrel is filled with oil to a depth of 20 cm. A steel marble of mass 0.1 kg is released from rest 60 cm above the top surface of the oil. Assuming that the oil exerts a constant force of 4 N on the marble, find the speed at which the marble hits the bottom of the barrel.[CO-2, L3]

2. A 40 kg child climbs up 4 metres up a flagpole and then slides to the bottom. If the child had a speed of 2.5 m/s at the bottom, what was the work done by friction? [CO-3, L3]

3. A sailor on board a ship of mass 2×10^6 kg moving at 0.3 m/s throws a heavy rope to a man standing on the wharf who drops the noose around a stout wooden peg. After the rope is taut, the ship is bought to rest, stretching the rope 0.5 m. Find the average pull sustained by the rope. [CO-2, L3]

4. A particle is acted upon by the force $\mathbf{F} = ax^4\,\hat{\imath} + bxy^2\,\hat{\jmath}$, where a, b are constants. Find the work done by this force along a straight line path from (2,4) to (4,8). [CO-2, L3]

5. Show that $\nabla \times \nabla V$ is zero for any scalar function V. [CO-5, L2]

6. Evaluate the curl of the function $\vec{F} = 6x^2\,z^2\hat{i} + 7xy\hat{j} + 15xz^2\hat{k}$ at the point (3,2,6). [CO-5, L2]

Section B : 10 marks each

7. A rocket of total mass m_0 in interplanetary space is moving at speed v_0. It has to speed up to $2v_0$. Find the amount of fuel that must be burnt. [CO-3,L3]

8. A block of mass 5 kg is on a horizontal surface. The coefficient of sliding friction is 0.5. The block has a speed of 10 m/s and is moving towards a stationary spring 3 metres away, which has a spring constant 4000 N/m. The block strikes the spring, compresses it, and bounces back. How far from the spring does the block comes to rest. [CO-2, L4]